

Start Here

ChatGPT edition

Choose an investigation, set the boundary, then make evidence

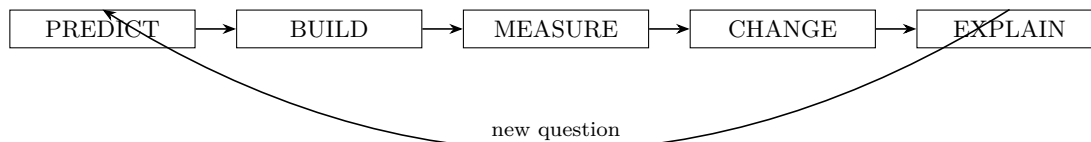
THE BOUNDARY IS PART OF THE EXPERIMENT

Every learner-built circuit in this set uses isolated battery power, **12 V DC maximum**. **STOP: no outlets, mains leads, wall adapters, large battery packs, large capacitors, ignition coils, or exposed high voltage.** An adult controls materials, checks batteries for damage or heat, and owns the decision to stop. An adult is *not* automatically qualified for exposed mains or high-voltage work; that work is outside this course.

Pick the question you want to answer

Investigation book	Question	You leave with
Charge & Field	How can an invisible interaction be mapped?	an evidence map, not a picture of “stuff”
Circuits by Evidence	Which model predicts brightness and current?	a measured V - I graph and power budget
Current Makes Fields	What changes an electromagnet’s effect?	a fair test of turns or current
Change Makes Current	Why does motion sometimes make voltage?	a sign-and-size induction record

The investigation loop



A result is not yet an explanation

Write: “When we changed _____, _____ changed from _____ to _____. Therefore _____, because _____.” Include units.

Roles that make independence possible

Learner	Chooses the prediction; wires only after the power source is disconnected; reads the diagram; records raw values; explains uncertainty.
Adult facilitator	Selects only the listed low-energy parts; inspects cells and leads; checks meter mode and jacks; stops heat, odor, swelling, damaged insulation, or repeated unexplained readings.
Nobody in this course	Opens an appliance; inserts anything into an outlet; measures household power; substitutes a wall adapter; or treats “off” as proof of deenergization.

BEFORE POWER IS CONNECTED

Circuit matches the diagram ☐ No direct battery short ☐ Meter mode and lead jacks checked ☐ Adult checks once ☐ Connect only long enough to measure ☐

Common kit

- holder for 2 AA cells; fresh alkaline cells
- 6 V lantern battery *or* 4-AA holder where specified
- insulated clip leads; pushbutton
- LEDs plus 330 Ω and 1 k Ω resistors
- 10, 47, 100, 220 and 1000 Ω resistors
- digital multimeter with intact, shrouded leads
- enamelled magnet wire; iron nail; compass
- strong permanent magnet; cardboard tube
- clear tape, index cards, ruler, graph paper

Evidence: OSHA S01–S03; NIST S04–S05; DOE S06–S07; CPSC S08–S09. See the provider-neutral electricity source register.

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